EVALUATION REPORT

# AptiGenie — Generative-AI Aptitude & Interview Prep Coach

An evaluation of the AptiGenie prototype's generated content, prompt-engineering design, and simulated learner performance system, against conceptual accuracy, curriculum alignment, personalization, reliability, fairness, and learning-effectiveness criteria — with recommendations for hallucination mitigation, data privacy, and ethical AI use.

| Prepared for | Subject | Date |
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| Prompt Engineering — CO2: Analyze | AptiGenie prototype v1 | 19 August 2026 |

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## 01 Executive Summary

AptiGenie is a prototype Generative-AI aptitude and interview-preparation coach built to the brief's specification: learner profiling, a personalized roadmap, concept-first topic teaching, adaptive practice, timed quizzes and mock tests, an auto-scored performance dashboard, and a reusable prompt-engineering layer covering role, structured, template, few-shot, chaining, and iterative-refinement techniques.

This report evaluates the prototype against the ten dimensions named in the brief — conceptual accuracy, question quality, solution correctness, curriculum alignment, personalization, consistency, explainability, reliability, fairness, and learning effectiveness — and sets out concrete strategies for minimizing hallucination, protecting learner data, and using AI ethically in this context.

The headline finding: the prototype's **static content is strong** — every formula and worked solution was hand-verified during authoring, and the system's explanations are transparent by design. Its **adaptive intelligence is currently rule-based rather than model-generated** — the roadmap, weak-area detection, and question feedback all run on deterministic logic that mirrors what a chained LLM pipeline would produce, documented explicitly in the app's Prompt Studio, but not yet wired to a live model. This is an appropriate and honest scope for a prototype; Section 9 sets out what live-API integration would require.

## 02 System Overview

AptiGenie is delivered as a single self-contained web application (no backend, no build step) covering thirteen aptitude topics across the Quantitative, Logical Reasoning, and Verbal Ability domains named in the brief — Number System, Percentage, Profit & Loss, Ratio & Proportion, Time & Work, Time-Speed-Distance, Simple & Compound Interest, Permutation & Combination, Probability, Blood Relations, Seating Arrangement, Coding-Decoding, and Verbal Ability. Data Interpretation, Puzzles, and Advanced Analytical Reasoning are scoped in the Learn view as "coming soon" rather than populated with unverified content — an explicit design choice discussed further in Section 8.

### Feature coverage against the brief

| BRIEF REQUIREMENT | IMPLEMENTATION |
|---|---|
| Personalized learner profile | Onboarding form (background, target, level, prep time, focus topics, goals), persisted locally |
| Personalized roadmap | `generateRoadmap()` — rule-based, weekly milestones paced to prep time and skill level |
| Concept-first teaching | Learn view: plain-language intro, formulas, one worked example, shortcuts, real-world use — per topic |
| Progressive practice with solutions | Practice view: adaptive difficulty (two correct → harder; one miss → easier), full step-by-step solution on every answer |
| Quizzes, section tests, mock tests | Quiz view: Topic Quiz (5Q/5min), Section Test (10Q/10min), Full Mock (26Q/20min), all timed and auto-scored |
| Scoring, accuracy, percentile, competency reports | Reports view: topic-wise accuracy chart, score trend, simulated percentile, weak-area recommendations, badges |

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|---|---|
| Reusable prompt workflows, multiple versions, comparison | Prompt Studio: five documented prompt templates (v1→v2/v3) with technique labels, rationale, and a live compose/regenerate demo |
| Downloadable preparation report | Print-formatted report (browser print / save-as-PDF) generated from live profile and progress data |

## 03 Prompt Engineering Approach

The brief asks for reusable prompt workflows applying role prompting, structured prompting, template prompting, few-shot prompting, prompt chaining, and iterative refinement, with multiple versions to compare. AptiGenie's Prompt Studio documents five such workflows, each mapped to the pipeline stage it would drive in a live deployment:

| STAGE | TECHNIQUE | WHAT IT PRODUCES |
|---|---|---|
| 1. Concept explanation | Role prompting + structured output template | The Learn-view card: intro, formulas, worked example, shortcuts, real-world use, as strict JSON |
| 2. Question generation | Few-shot prompting + prompt chaining | New MCQs grounded in Stage 1's approved formula list, matching a worked few-shot format |
| 3. Answer feedback | Structured + role prompting | The one-line coaching message shown after each answer, with solution steps passed through unchanged |
| 4. Roadmap generation | Multi-stage prompt chaining | Topic selection → weekly pacing → adaptive revision insert, each stage validated before the next consumes it |
| 5. Revision recommendation | Iterative refinement (v1→v3) | Weak-topic flags, refined across three iterations to add a minimum-attempts guard |

Each template in the app ships with its rejected earlier version and a one-paragraph rationale for the change — for example, Stage 5's revision recommender went from an open-ended v1 ("Which topics is the student weak in?") to a v3 that refuses to flag a topic from a single data point, a fix made after reviewing v2's output against sample quiz logs. This version history is the artifact the brief asks for under "prompt customization, regeneration, comparison, and optimization" — the Prompt Studio's live playground lets a reader fill Stage 2's template with real topic/difficulty/count inputs and cycle through sample outputs, standing in for what a Regenerate button against a live API would do.

> **Scope note.** In this prototype, the content these prompts describe was generated once, by hand, following the exact template — then verified — rather than produced by live calls to a model such as Claude at runtime. The templates are production-ready; the missing piece for a v2 build is an API integration layer that sends them to a Generative AI endpoint and validates the structured response before it reaches a learner. This is scoped in Section 9.

## 04 Evaluation Methodology

Each of the ten dimensions named in the brief was scored on a 1–5 rubric by manual review against three references: standard placement-aptitude texts (formula and method cross-check), the topic list and question style typical of campus recruitment tests (TCS NQT, Infosys, capgemini-style papers), and the WCAG-aligned accessibility practice of never encoding meaning by color alone. Scores reflect the *current prototype build*, not a theoretical ceiling for the architecture — several dimensions are marked down specifically because they depend on a live model connection this build does not yet have.

| BAND | SCORE RANGE | MEANING |
|---|---|---|
| STRONG | 4.0 – 5.0 | Meets the criterion with verifiable evidence in the shipped build |
| MODERATE | 3.0 – 3.9 | Partially met; a specific, named gap remains |
| GAP | Below 3.0 | Not yet met in this build; requires further work before production use |

## 05 Evaluation Results

| DIMENSION | SCORE | EVIDENCE & GAP |
|---|---|---|
| Conceptual accuracy | 4.5 / 5 STRONG | All 39 formulas across 13 topics were cross-checked against standard aptitude references and verified by substitution during authoring (see Section 6). |
| Question quality | 4.0 / 5 STRONG | 50 questions span easy/medium/hard with plausible distractors in placement-exam phrasing; bank size per topic (3–5 Qs) is too small to prevent repeats across repeated practice sessions. |
| Solution correctness | 4.5 / 5 STRONG | Every answer was independently recomputed by hand during authoring; no automated symbolic-solver cross-check exists yet, so correctness currently rests on single-author manual verification. |
| Curriculum alignment | 3.5 / 5 MODERATE | 13 of the brief's ~17 named domains are fully populated; Data Interpretation, Puzzles, and Advanced Analytical Reasoning are marked "coming soon" rather than shipped with unverified content. |
| Personalization | 3.0 / 5 MODERATE | Profile drives topic order, pacing, and target accuracy; personalization is rule-based on profile fields, not yet adaptive to individual answer patterns beyond the difficulty ladder and >2-attempt weak-area rule. |
| Consistency | 3.5 / 5 MODERATE | Rule-based logic is perfectly reproducible by construction; this does not demonstrate the run-to-run consistency of an actual LLM |

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| | | generation pipeline, which has not yet been measured. |
| Explainability | 4.5 / 5 STRONG | Every question ships a step-by-step derivation; every recommendation states the exact threshold rule that produced it (e.g. the ≥2-attempt guard). |
| Reliability | 3.5 / 5 MODERATE | Fully client-side and offline-capable after load; but progress lives only in one browser's local storage with no sync or backup, and percentile is a simulated statistic, not a real benchmark. |
| Fairness | 3.0 / 5 MODERATE | Accuracy is always paired with a numeric label, never color alone; the question set has not been formally audited for cultural or linguistic bias beyond its intentionally India-focused placement context. |
| Learning effectiveness | 3.0 / 5 MODERATE | Adaptive difficulty and revision recommendations follow evidence-based tutoring patterns; no real-learner study has yet measured their effect on outcomes. |

**Overall: 3.7 / 5** — the static, author-verified layer (accuracy, correctness, explainability) is consistently strong; every dimension that depends on either a live model connection or real learner data is consistently moderate, which is the expected shape for a prototype at this stage rather than a random spread of weaknesses.

## 06 Content Validation Process

Every formula and worked example in AptiGenie's thirteen topics was validated against standard aptitude preparation references (the RS Aggarwal / R.S. Agarwal school of quantitative-aptitude and reasoning texts widely used for Indian campus placement prep) and cross-checked by independent recomputation — not merely re-stated from a source. For example, the 24-factor count for 360 was verified via its prime factorization ($2^3 \times 3^2, 5^1 \rightarrow 4\times3\times2$), and the Blood-Relations puzzles were verified by drawing out the family tree generation-by-generation rather than accepted on pattern-match alone. All 50 questions carry a fully worked, human-checkable solution rather than an answer key alone, which is itself a validation control: an unverifiable final answer with no derivation is a common source of undetected error in AI-generated question banks, and this design makes every answer auditable by a reviewer or student.

This manual, single-pass validation is appropriate for a prototype of this size but does not scale — Section 9 recommends the automated and multi-reviewer controls a production version would need.

## 07 Risk & Mitigation Strategy

The brief asks specifically for strategies to minimize AI hallucination, protect learner data privacy, and ensure ethical AI use. Each is addressed below as it applies to AptiGenie's architecture.

`Hallucination` **Minimizing fabricated formulas, wrong answers, and invented topics**

- **Grounded generation over free generation.** Stage 2's question-generation prompt (Section 3) explicitly chains in Stage 1's approved formula list and forbids inventing new ones — the model is constrained to a verified formula set rather than free to derive its own.
- **Structured output, not prose.** Every generation stage returns a fixed JSON schema; a response that fails to parse or is missing a required field is a detectable failure rather than a silently wrong answer reaching a learner.
- **Independent answer verification.** Before any generated question reaches a learner, its stated answer should be re-derived by an independent method — a symbolic/CAS solver for algebraic topics, a small test-oracle script for numeric ones — and rejected on mismatch, rather than trusted on the model's own "steps" narrative.
- **Human-in-the-loop review for the seed bank.** As implemented here, the current 50-question bank was authored against the template and hand-verified rather than machine-generated at runtime — the same review gate should sit between any live-generated batch and publication.
- **Confidence flagging.** A production pipeline should have the model self-report low confidence on a generated item (or route topics it has weak formula coverage for) so uncertain content is queued for review instead of shipped silently.

`Privacy` **Protecting learner data**

- **Data minimization by default.** The prototype collects only what the roadmap needs — name, education level, target, skill level, prep time, focus topics — and stores it in the browser's own local storage, never transmitted to a server in this build.
- **Local-first as a privacy control, not just an implementation shortcut.** Because there is no backend, there is no learner database to breach; a production version that adds sync should keep this local-first option available rather than making cloud storage mandatory.
- **Minimal necessary context to any LLM call.** When wired to a live API, prompts should pass profile fields needed for personalization (level, target, topic) and never the learner's name or other identifying detail into a third-party model call.
- **Explicit retention and consent.** A production deployment needs a stated retention period, a visible export/delete-my-data control, and compliance review against India's Digital Personal Data Protection Act, 2023 (and GDPR if serving EU users) before it can legally operate on named minors' or students' data.
- **Anonymized aggregate analytics only.** Any cross-learner analytics (e.g. "average accuracy on Time & Work") should be computed on de-identified, aggregated data, never on raw per-learner logs.

Ethics **Ensuring ethical AI use**

- **Transparency about what is AI-generated.** The Prompt Studio makes the generation process visible rather than presenting AI output as an anonymous authority — a design principle that should extend into the learner-facing app itself (e.g. a visible "AI-generated, human-reviewed" tag on new content).
- **Guarding against over-reliance in mock tests.** A mock test's purpose is to simulate real exam conditions; its scoring and percentile must be clearly labeled as an estimate (as done in this build's Reports view) so learners do not mistake a simulated cohort comparison for a real one.
- **Bias auditing.** The question set's names, contexts and cultural references should be periodically reviewed for unintended bias — this build is intentionally India/placement-context-specific, which is appropriate for its stated audience but should be a documented scope decision, not an unexamined default.

- **No dark patterns in gamification.** Badges and streaks (Section 2) are designed to reflect genuine progress thresholds (e.g. ≥2 attempts before a weak-area flag) rather than manufactured urgency; this principle should hold as gamification features expand.
- **Accessible-by-default status signaling.** Every accuracy band in the Reports view pairs color with a numeric label, so the system's own "judgment" of a learner's performance is never communicated through color alone.

## 08 Limitations of the Prototype

- **No live Generative-AI API call.** All prompt templates in Section 3 are production-ready but not yet wired to an actual model endpoint; content was generated once, by hand, following the templates, then verified — see the scope note in Section 3.
- **Small question bank.** 3–5 questions per topic will repeat within a handful of practice sessions; a production bank needs continuous, reviewed generation to stay ahead of repeat exposure.
- **Simulated percentile.** The percentile shown in Reports is computed against an assumed peer-cohort distribution (mean 62%, s.d. 16), not measured against real test-takers — clearly labeled as such in the UI, but not a substitute for real benchmark data.
- **Single-device persistence.** Progress lives in one browser's local storage with no account system, sync, or backup.
- **Three syllabus domains unpopulated.** Data Interpretation, Puzzles, and Advanced Analytical Reasoning are scoped but not built, and are shown as locked rather than filled with placeholder content.
- **No real-learner validation.** Learning-effectiveness claims (Section 5) are grounded in established tutoring-design patterns, not an outcome study on actual students.

## 09 Recommendations & Next Steps

To move from this prototype toward a production-ready system, in priority order:

1. **Wire the Prompt Studio templates to a live Generative AI API** (e.g. the Claude API), with the structured-output schemas in Section 3 enforced at the integration layer so a malformed or ungrounded response is rejected before reaching a learner.
2. **Add an automated answer-verification step** between generation and publication — a symbolic solver for algebraic topics, a rule-checker for logic-puzzle topics — so no generated question ships on the model's self-reported correctness alone.
3. **Expand the question bank per topic** to the 30–50 range needed to support spaced repetition without noticeable repeats, generated in reviewed batches via the Stage 2 template.
4. **Build the three remaining syllabus domains** (Data Interpretation, Puzzles, Advanced Analytical Reasoning) using the same validate-before-ship process documented in Section 6.
5. **Replace the simulated percentile with a real cohort benchmark** once enough real learners have used the mock-test feature to compute one honestly.
6. **Add account-based sync** behind an explicit, revocable consent flow, keeping the current local-only mode available as a privacy-preserving default.
7. **Commission a small real-learner pilot** (a single cohort across one prep cycle) to test the learning-effectiveness claims in Section 5 against actual outcome data rather than design-pattern precedent alone.

## 10 Conclusion

AptiGenie meets the brief's functional specification end-to-end — profiling, roadmap, concept teaching, adaptive practice, timed assessment, performance reporting, and a documented, versioned prompt-engineering layer — and does so with content that has been verified rather than merely generated. Its evaluation profile is coherent rather than scattered: dimensions resting on hand-authored, human-checkable content score strong, while every dimension that depends on a live model connection or real learner data scores moderate and is accompanied by a named, addressable gap. That is the appropriate shape for a prototype, and Section 9's roadmap describes exactly what closes each gap on the path to a production deployment.